

COURSE TITLE : CAD LAB
COURSE CODE : 3079
COURSE CATEGORY : B
PERIODS/ WEEK : 6
PERIODS/ SEMESTER : 90
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to Auto Cad 2D	24
2	Auto CAD with 2D & Isometric Drawing	27
3	PIPE JOINTS& COUPLINGS in 2D	21
4	Assembly and Detail Drawings	18
TOTAL		90

COURSE OUTCOME

SL.NO.	GO	STUDENT WILL BE ABLE TO
1	1	Familiarize with system requirements for CAD.
	2	Familiarize with AutoCAD interface, different entities and editing commands.
	3	Set up a drafting with different layers.
2	1	Familiarize with texting and various methods of dimensioning.
	2	Familiarize with important fastening devices and their details.
	3	Create isometric objects and isometric dimensioning.
3	1	Familiarize with various pipe joints.
	2	Familiarize with various couplings.
	3	Expertise in 2D drafting
4	1	Create a block
	2	Insert a Block
	3	Assign and use attributes
	4	Set up assembly of components drawn separately.

CONTENT DETAILS

MODULE- I

History of computers- elements of computer-types of computers.

Introduction to Computer Aided Drafting- Definition of CAD- Applications and advantages of CAD.

Getting started with Auto CAD Software.

Main menu- Auto CAD editor screen – command entry-pull down menu.

Utility commands

Help, End, Quit, Save and Limits.

Drawing entities

Point, Line (Absolute co-ordinates, relative co-ordinates, polar co-ordinates),

PLINE, Rectangle, Circle, Ellipse, Arc and Polygon.

Auto CAD commands

Setting of drawing aids

ORTHO, SNAP and GRID commands. Use of function keys, starting a new drawing, saving a file, setting units for Engineering drawing.

Starting AutoCAD.

Selecting commands in AutoCAD - Key board, Tool bar, Screen menu, Pull down menu and Digitizing tablet.

Editing-Types of editing commands – ERASE, UNDO, REDO, SELECT,CHANGE,COPY ,ARRAY, OFFSET , MIRROR, MOVE, ROTATE, , TRIM,EXTENT, JOIN, BREAK, HATCH, CHAMFER and FILLET.

Layers-Layer property manager - creation of layers.

MODULE- II

Basic dimensioning

Methods of dimensioning - Unilateral dimensioning and Aligned dimensioning. Different types of dimensioning commands – Drawing commands(LINEAR, ANGULAR, DIAMETER,ORDINATE), STYLE commands(OVERRIDE, RESTORE, SAVE, VAIRABLES, STYLE), EDITTING commands (HOMETEXT, NEWTEXT, OBLIQUE,TEDIT, UPDATE) and UTILITY commands(EXIT, LEADER, REDRAW, STATUS, UNDO)

Developing simple orthographic views and dimension it with text.

Developing detailed orthographic views with all features (bolts and nuts, Locking nuts, rivet heads, different types of bolted joints and different types of riveted joints.

Isometric Drawing

Isometric axes – isometric planes-setting isometric grid and snap – isoplane – isometric Cube and Isometric Circles – isometric text.

MODULE- III

PIPE JOINTS& COUPLINGS in 2D

Pipe Joints

Flanged, Socket and Spigot joint, Screwed union joint, Integral flanged, screwed flanged.

Couplings

Muff Coupling (Solid & Split type), Flanged Coupling (Protected &Un-protected type) and Flexible Coupling.

MODULE- IV

Block

Creating a block – insert a block – presetting the insertion point , scale or rotation

Attributes

Creating block attributes – attribute modes (invisible, constant, verify and preset)

Assembly and Detail Drawings in 2D

Need and functions of assembly and detailed drawings . Steps in preparing assembly and detailed drawings. Cotter joint, Knuckle joint, Foot step bearings, Plummer block and Stuffing box.

REFERENCES

- | | | | |
|------------------|--------------------------------|------------------------------------|----------------|
| Cheryl R. Shrock | - Advanced AutoCAD2013 | - New Age international Publishers | - First |
| | Exercise workbook for Windows | | |
| K. Venugopal | - Engineering Drawing and | - New Age international Publishers | - Fourth |
| | Graphics + AutoCAD | | |
| P. Radhakrishnan | | | |
| S. Subramanyan | - CAD/CAM/CIM | - New Age international Publishers | - Third |
| V. Raju | | | |
| P I. Varghese | - Machine Drawing | - VIP Publishers | - Twenty Sixth |
| Don W Green | | | |
| Robert Perry | - Perry's Chemical Engineering | - Mc Graw-Hill | - Eighth |
| | Hand Book | | |